# PlantDescriptionEngine

## System Description

**Abstract**
This document describes the PlantDescriptionEngine support system of the Eclipse Arrowhead framework (documentation edition 5.3.0), as implemented in `ah5-PlantDescriptionEngine-java-spring` of the primary Java source tree (`Eclipse-Arrowhead-java`) and deployed with the standalone Compose definition of that folder.

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# 1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

# 2 Overview

This document describes the PlantDescriptionEngine system. The PlantDescriptionEngine keeps the cloud's orchestration consistent with an engineered plant description: it turns the active description's connections into a managed snapshot of DSO provider rules, reconciles them periodically and raises alarms where the running cloud differs.

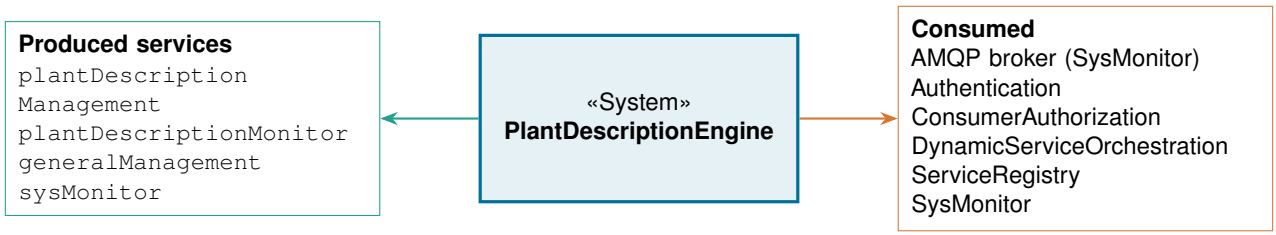



Figure 1: SysML-style block view of PlantDescriptionEngine with its produced and consumed services.

The rest of this document is organized as follows. In Section 2.1, we reference major prior art capabilities of the system. In Section 2.2, we describe the intended usage of the system. In Section 2.3, we describe fundamental properties provided by the system. In Section 2.4, we describe delimitations of capabilities of the system. In Section 3, we describe the abstract service functions consumed or produced by the system. In Section 4, we describe the security capabilities of the system.

## 2.1 Significant Prior Art

Successor of the Arrowhead 4.6 PlantDescriptionEngine.

## 2.2 How This System Is Meant to Be Used

Engineers (or LocalCloudManager's PD view) maintain plant descriptions; the PDE atomically replaces the active-plant rules at DynamicServiceOrchestration and reports alarms through `plantDescriptionMonitor`.

## 2.3 System functionalities and properties

### 2.3.1 Functional properties of the system

- Validation of plant descriptions and includes.
- Matching of plant systems to registered systems (name and metadata).
- Atomic snapshot synchronization with DynamicServiceOrchestration.
- Alarms from registry state and SysMonitor history.

### 2.3.2 Configuration of system properties

The system is configured with Spring properties (`application.properties`, overridden by environment variables in the container deployment). The properties that shape its behaviour are:

| Property | Default | Meaning |
|---|---|---|
| `server.port` | 8432 | HTTPS port. |
| `authentication.policy` | declared | Standalone deployment: `outsourced` with Authentication identity tokens and mutual TLS. |
| `management.policy` / `management.whitelist` | authorization | Access to the management services. |
| `enable.blacklist.filter` / `force.blacklist.filter` | false / true | Blacklist check of requesters. |
| `service.authorization.enabled` | false | Per-call ConsumerAuthorization check (`authorization.verify`, cached 30 s, fail closed); standalone deployment: true. |
| `plant.sync.interval` | 15000 | Synchronization interval (ms). |
| `plant.rule.default.priority` | 1 | Priority of connections without one. |
| `plant.monitoring.enabled` / `plant.monitoring.stale.after.seconds` | true / 120 | Monitoring alarms. |
| `plant.alarm.ignored.systems` | framework systems | Exempt from `SYSTEM_NOT_IN_DESCRIPTION`. |
| `plant.description.max.size` | 200000 | Maximum JSON size. |
| `normalization.mode` | extended | Name normalization. |
| `sysmonitor.enabled`, `sysmonitor.broker.*`, `sysmonitor.ssl.*` | true, amqp-broker:5671, server.ssl stores | Native `sysMonitor` publisher of the common library. |
| `arrowhead.ssl.crl-file` | (unset) | CRL checked for every client certificate (standalone deployment: `/crl/ca.crl`). |

Secrets (database, keystore and identity passwords, secret keys) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets under `/run/secrets`.

The standalone deployment of this folder uses `config/application.properties` (copied into the image) with `server.port`=8432, `authentication.policy=outsourced`, `service.interface.policy=CERT_AUTH`, `management.policy=sysop-only`, `service.authorization.enabled=true`, `mqtt.api.enabled=false`, `arrowhead.ssl.crl-file=/crl/ca.crl`; secrets are read from Compose file secrets under `/run/secrets` (`spring.config.import=configtree:/run/secrets/`). `enable.blacklist.filter` is not enabled there, so incoming requests are not checked against the Blacklist; blacklisted providers are still removed by DynamicServiceOrchestration.

### 2.3.3 Data stored by the system

Plant descriptions, alarms and log entries (database `ah_plantdescriptionengine`).

### 2.3.4 Non functional properties

`echo` is `degraded` while synchronization fails or alarms are active.

### 2.3.5 Stateful or stateless

Stateful: descriptions and alarms are persistent.

## 2.4 Important Delimitations

Requires the local DSO plant-management extension and schema migration. Managed consumer/service scopes restrict candidates; empty scopes deny unrelated or inter-cloud fallback. No active plant clears the snapshot.

# 3 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

## 3.1 Produced services

| Service | Version | Interfaces | SD | IDD |
|---|---|---|---|---|
| `plantDescriptionManagement` | 1.0.0 | HTTP(S) | [1] | [2] |
| `plantDescriptionMonitor` | 1.0.0 | HTTP(S) | [3] | [4] |
| `generalManagement` | 1.0.0 | HTTP(S) | [5] | [6] |
| `sysMonitor` | 1.0.0 | AMQP(S) | [7] | [8] |

The HTTP interfaces are always offered and the AMQP `sysMonitor` interface with `sysmonitor.enabled=true`.

## 3.2 Consumed services

| Provider | Service | Operations | Use | Condition |
|---|---|---|---|---|
| ServiceRegistry | systemDiscovery, serviceDiscovery | register, revoke, lookup | Registration of the system and its services; lookup of the bootstrap services. | always |
| ServiceRegistry | serviceRegistry Management | system-query, service-query | Matching of plant systems. | management whitelist |
| Authentication | identity | identity-login, identity-verify | Own identity token; verification of requester tokens. | `authentication.policy=outsourced` |
| Consumer Authorization | authorization | verify | Per-call authorization of its services. | `service.authorization.enabled=true` |
| DynamicService Orchestration | serviceOrchestration PlantManagement | query, replace | Atomic active-plant snapshot; DSO still checks authorization, locks, blacklist and interfaces. | scoped operation grants |
| SysMonitor | sysMonitorHistory | list-systems, latest | Liveness of plant systems. | `plant.monitoring.enabled=true` |
| AMQP broker (SysMonitor) | sysMonitor (publisher) | echo, power, status | Own observations. | `sysmonitor.enabled=true` |

Except for the bootstrap services, which are looked up in the ServiceRegistry, the system reaches other systems' services through DynamicServiceOrchestration (`http.collector.mode=ORCHESTRATION`, results cached for 300 s); every such call needs a ConsumerAuthorization grant.

# 4 Security

The system can run in an insecure mode (`server.ssl.enabled=false`, plain HTTP, identity declared by the requester) for development, and in the Arrowhead secure mode with HTTPS and X.509 certificates following the Arrowhead certificate profile (`<System>.<Cloud>.<Organization>.arrowhead.eu`). Certificates not issued under the cloud CA are refused by the TLS handshake. The standalone deployment of this folder (`config/application.properties` and `compose.yaml`) runs the secure mode.

## 4.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mandatory client certificates; MQTT over TLS; AMQP over TLS for `sysMonitor`.
- **Data protection:** transport encryption by TLS; the database connection uses TLS with server verification in the standalone deployment. Secrets are kept out of the container environment (Compose file secrets).
- **System authentication:** by the authentication policy: `certificate`, `outsourced` (identity token verified with Authentication) or `declared` (insecure); the standalone deployment uses `outsourced`. An identity token is accepted only together with the client certificate of the system it was issued to (token-to-certificate binding), and every client certificate is checked against the cloud CA's CRL (fail closed).
- **Authorization of produced services:** management services are protected by the management filter (`management.policy`: `SYSOP_ONLY`, `WHITELIST` or `AUTHORIZATION`); with `service.authorization.enabled=true` ConsumerAuthorization decides each call; an optional blacklist filter refuses blacklisted requesters. Sysop identities bypass the management filter.
- Standalone deployment: `management.policy=sysop-only`. The plant-description and SoS views of LocalCloudManager (not a sysop) need `management.policy=whitelist` with `management.whitelist=LocalCloudManager`, as in the earlier combined test-cloud deployment.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

# 5 References

[1] J. Delsing, "plantDescriptionManagement – Service Description (PlantDescriptionEngine)," Eclipse Arrowhead, SD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-plantDescriptionManagement-SD.pdf.

[2] ——, "plantDescriptionManagement HTTP/TLS/JSON – Interface Design Description (PlantDescriptionEngine)," Eclipse Arrowhead, IDD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-plantDescriptionManagement-HTTP-IDD.pdf.

[3] ——, "plantDescriptionMonitor – Service Description (PlantDescriptionEngine)," Eclipse Arrowhead, SD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-plantDescriptionMonitor-SD.pdf.

[4] ——, "plantDescriptionMonitor HTTP/TLS/JSON – Interface Design Description (PlantDescriptionEngine)," Eclipse Arrowhead, IDD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-plantDescriptionMonitor-HTTP-IDD.pdf.

[5] ——, "generalManagement – Service Description (PlantDescriptionEngine)," Eclipse Arrowhead, SD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-generalManagement-SD.pdf.

[6] ——, "generalManagement HTTP/TLS/JSON – Interface Design Description (PlantDescriptionEngine)," Eclipse Arrowhead, IDD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-generalManagement-HTTP-IDD.pdf.

[7] ——, "sysMonitor – Service Description (PlantDescriptionEngine)," Eclipse Arrowhead, SD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-sysMonitor-SD.pdf.

[8] ——, "sysMonitor AMQP/TLS/JSON – Interface Design Description (PlantDescriptionEngine)," Eclipse Arrowhead, IDD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-sysMonitor-AMQP-IDD.pdf.

# 6 Revision History

## 6.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-23 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |
| 2 | 2026-10-30 | 5.2.2 | Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated. | Claude Opus 5.5 (draft) |
| 3 | 2026-11-09 | 5.3.0 | New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained. | Documentation maintenance |

## 6.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.3.0 | |